

Web Development

Dr. rer. nat. Isara Anantavasilp

Lab 2: CSS

Project: Thailand School and Hospital Explorer

In Lab 1, you created the HTML structure of the Thailand School and Hospital Explorer. In this lab, you will continue the same project and add CSS to control the page's appearance, spacing, and layout. More precisely, you will:

- move presentation into an external CSS file
- style text, colors, backgrounds, borders, and spacing
- apply the CSS box model
- use selectors to target specific elements
- use Flexbox to arrange the navigation menu
- style the facility table and other page sections.

The HTML content and semantic structure from Lab 1 should remain the foundation of your project.

Objectives

Continue the website from Lab 1 and improve its appearance and layout using CSS.

- Connect an external stylesheet to an HTML document
- Use element, class, ID, and descendant selectors
- Apply fonts, colors, backgrounds, borders, margins, and padding
- Explain and apply the CSS box model
- Use `box-sizing: border-box`
- Use Flexbox to arrange elements in one dimension
- Inspect and modify CSS using browser DevTools

Assignments

1. Continue from Lab 1

Create a new project folder named:

Open the project folder that you created in Lab 1.

Add one new file:

```
thailand-facility-explorer/  
├── index.html  
├── styles.css   
└── images/  
    └── Thailand_adm_location_map.svg
```

2. Link an External Stylesheet

Inside the <head> element of index.html, link the new CSS file:

```
<link rel="stylesheet" href="styles.css">
```

***** All required styling for this lab should be written in styles.css. Do not use inline styles for the required work. *****

Reload the page after adding a simple CSS rule to confirm that the stylesheet is connected correctly.

3. Create the Global Page Style

Begin styles.css with a common box-sizing rule:

```
* {  
    box-sizing: border-box;  
}
```

Then style the <body>. Your page must set at least:

- a font family
- a text color
- a background color
- the body margin

Example:

```
body {  
    font-family: Arial, Helvetica, sans-serif;  
    margin: 0;  
    color: #333;  
    background-color: #f5f5f5;  
}
```

You may choose different colors and fonts. Keep the design readable and consistent.

4. Style the Header

Style the <header> so that it is visually different from the main content.

The header style must use:

- background-color
- color
- padding

Also style the main <h1>. You may change its size, color, or spacing.

5. Create a Flexbox

The navigation list from Lab 1 is vertical by default. Use Flexbox to display the navigation items horizontally.

Example:

```
nav ul {  
    display: flex;  
    gap: 20px;  
    list-style: none;  
    padding: 0;  
}
```

In this example:

- is the flex container
- elements are the flex items
- gap controls the space between the items.

Style the navigation links so that they are easy to read against the header background.

6. Apply the Box Model to the Main Content

Give the main content a maximum width and center it on the page.

Example:

```
main {  
    max-width: 1000px;  
    margin: 0 auto;  
    padding: 20px;  
}
```

Then style each <section> as a visible content box. Each section must use:

- margin
- padding
- border
- background-color.

Use DevTools to inspect one section and identify its content, padding, border, and margin.

7. Style the Overview Image

Remove the width attribute from the Thailand map image if you used one in Lab 1. Control its size using CSS instead.

Example:

```
#overview img {  
    width: 500px;  
    max-width: 100%;  
}
```

The image must remain inside the images folder and continue to use a relative path in HTML.

8. Style the Schools and Hospitals

Improve the readability of the school and hospital sections.

Use CSS to style at least two of the following:

- school list spacing
- hospital description spacing
- section headings
- borders or backgrounds
- a class that can be reused on more than one element.

Your stylesheet must contain at least one class selector. Add an appropriate class attribute to the HTML where needed.

9. Style the Facility Summary Table

The browser does not draw visible borders around table cells by default.

Add table styling using CSS.

Your table must include:

- collapsed borders
- visible borders around `<th>` and `<td>`
- padding inside cells
- a different style for the table header.

Example:

```
table {
    width: 100%;
    border-collapse: collapse;
}

th,
td {
    border: 1px solid #999;
    padding: 10px;
    text-align: left;
}

th {
    background-color: #1f4e79;
    color: white;
}
```

10. Style the Footer and Inspect the Final Page

Style the footer so that it is visually separated from the main content.

The footer must use at least:

- background-color
- color
- padding
- text-align

Open browser DevTools and inspect the final page.

Check that:

- styles.css is loaded without errors
- Flexbox is active on the navigation list
- The box model is visible for a section
- The page remains readable when the browser window is made narrower.

Checklist

Before submitting, confirm that your webpage:

- ☐ uses an external styles.css file linked correctly from index.html
- ☐ uses a global font, text color, and background style
- ☐ uses box-sizing: border-box
- ☐ styles the header using color, background, and padding
- ☐ uses Flexbox to display navigation items horizontally
- ☐ uses gap to create spacing between navigation items
- ☐ styles sections using margin, padding, border, and background
- ☐ styles the Thailand map image using CSS rather than an HTML width attribute
- ☐ styles the facility summary table with visible borders and cell padding
- ☐ styles the footer and maintains a consistent, readable design

Each checkpoint is worth 2 points.